

Emily Morazan

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Education

California State University, Northridge

B.S. in Computer Science, Minor in Mathematics

August 2022 – Current

GPA: 3.4

Skills

- Programming Languages: C, Java, Python, JavaScript, MIPS, HTML, CSS, React, Flutter, Flask
- Skills: Microsoft Office, Incident Response, Documentation, Data Structures, time management, GitHub, Machine Learning, and Embedded Apps,
- Languages: English, Spanish, French

Experience

Computer Science Tutor, College of Engineering and Computer Science

January 2026 – May 2026

Instructional Student Assistant | Tutored in Data Structures, OOP, algorithms, and IT

Los Angeles, CA

- Guided students in applying theoretical concepts to practical coding problems
- Collaborated with faculty and staff to align tutoring support with course objectives
- Provided one-on-one and small group tutoring sessions to support diverse learning styles

Student Assistant, College of Engineering and Computer Science

January 2024 – July 2025

Computer Science Student Assistant | Skilled in Microsoft Office and Administrative Coordination

Los Angeles, CA

- Provide information about Academic Enrichment Center resources and tutoring schedules to students in Engineering and Computer Science.
- Maintained organization of tutor schedules and student data using Microsoft Office, emphasizing accuracy and responsible handling of sensitive information.

M-STEP Tutor for STEM, CSUN

June 2024 – August 2024

Instructional Student Assistant | Tutored in Calculus, Geometry, Algebra, Pre-Calculus, and Linear Algebra

Los Angeles, CA

- Collaborate with faculty for proctor examinations and assessments
- Conduct tutoring sessions in a range of mathematical topics, including algebra, geometry, and calculus.
- Maintain accurate records of student performance and communicate regularly with Program coordinator, professor, and supervisor about progress.

Instructional Student Assistant, College of Mathematics

June 2023 – December 2023

Mathematic Tutor | Tutored in Calculus, Geometry, Algebra, and Pre-Calculus

Los Angeles, CA

- Assists over 100 College of Science and Mathematics students via the Mathematics Tutoring Center.
- Collaborate with faculty for proctoring assessments and facilitating testing activities.
- Provide feedback and support to help students set and achieve academic goals.

Projects

Network Traffic Analyzer & Firewall Web Application:

- Developed a full-stack web application to monitor and display real-time network activity, including incoming/outgoing connections, system information, and network speed.
- Implemented a simple firewall with connection-blocking rules and a user-friendly interface to visualize network traffic and system performance.
- Utilized tools such as **Scapy**, **Flask**, and **SQLite** for packet sniffing, data logging, and rule management.

Gamified Habit Tracker for ADHD

- **Won 1st** place at the **SS12 Hackathon**. (Flutter, Dart) Mentored teammates in debugging and troubleshooting component issues.
- Led a team of 6 to develop a gamified habit tracker for people with ADHD.
- Developed and implemented key components, including a task list, reward system, and notifications.

CodeReady — Interactive DSA Visualizer & AI Tutor | **Full-Stack Developer & UI/UX Lead**

- Real-time data-structure ops, step-through sorting visualizations with metrics, plus progress tracking, session restore, and leaderboards.
- Built React 19 front-end with a custom animation engine for step-through visualizations (Bubble/Quick/Merge Sort), real-time metrics, and complexity callouts.
- Integrated a context-aware AI tutor (Groq API) that adapts guidance to the user's current page/state using targeted prompt engineering.
- React 19, JavaScript, HTML/CSS (advanced animations, responsive design), Node.js, Express, Groq API, REST, LocalStorage, State Management.

Interactive Neural Network Playground

- Designed and built an **interactive neural network playground** that combines model training, real-time visualization, and explainability in a single interface.
- Developed a **React-based frontend** with **D3.js/Plotly** to display live training metrics (loss, accuracy, confusion matrix) and model internals.
- Focused on user-friendly interface design for complex technical workflows using **TypeScript**, **React**, HTML/CSS, and **JavaScript**

SignFlow — Sign Language to Text (Flutter + MediaPipe)

- Built an accessibility app translating ASL hand gestures into text
- Integrated MediaPipe hand tracking (web) with custom gesture rules
- Connected Flutter to JS via dart: js_interop for real-time detection

Extracurriculars

CSUN Layer 8 CCDC Team

August 2025 – May 15

Social Media Chair for American Society of Civil Engineers ASCE

August 2025 – May 15